BRAINHACK 2026

RNN Night Fever

Building the networks and the training pipeline that let RNNs replay the day while they sleep.

Perception **Hippocampus** **Cortex / PFC**

Lead: Alejandro Tabas

Team: Yasin Farshchitabrizi | Outhmane Rassili | Jacy To | Ane Urkola | Nursena Arslan | Yayu Yao | Nirmitee Mulay

What happens to memories during sleep?

Previously experienced neural activity can be reactivated during sleep.

DAY · EXPERIENCE

A sequence is experienced

Neural patterns are encoded

SLEEP · REACTIVATION

Related patterns can reappear

This phenomenon is called replay

Relevant evidence: Wilson & McNaughton (1994); Lee & Wilson (2002)

Two learning systems, two learning speeds

Complementary Learning Systems (CLS) theory

FAST

Hippocampus

Rapid learning of specific experiences

↔

SLOW

Neocortex

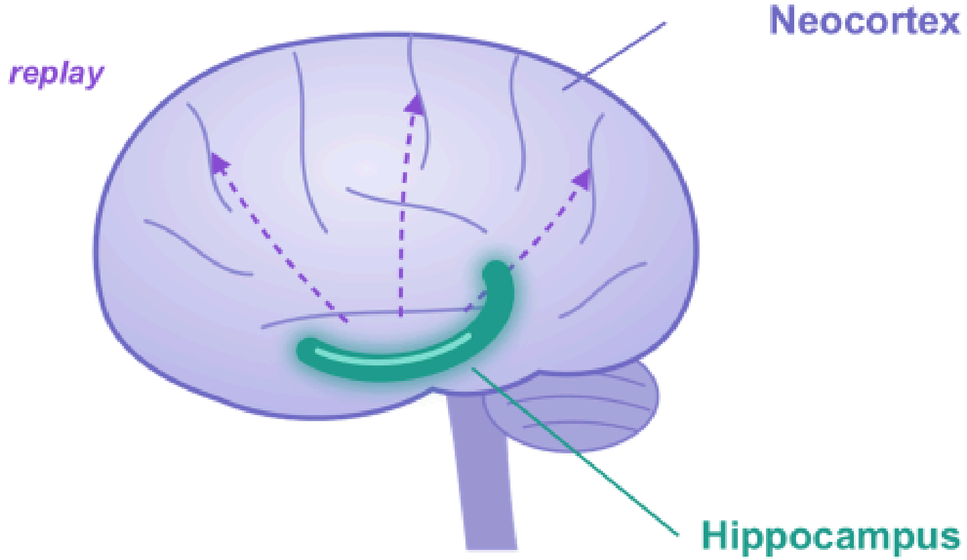
Gradual learning of regularities

Fast learning and slow learning are complementary — not identical.

McClelland, McNaughton & O'Reilly (1995); Schapiro et al. (2017)

How the hippocampus and neocortex work together

Learn fast during the day, consolidate slowly during sleep.



DAY · ENCODE

Hippocampus

Quickly stores each new experience as a separate memory (fast, small capacity)

SLEEP · REPLAY

Replay

Recent experiences are reactivated many times, faster than in real time

LONG TERM · LEARN

Neocortex

Slowly extracts regularities and builds an internal model of the world (large capacity)

Outcome: the hippocampus is freed for new learning, and the cortex learns without overwriting old knowledge.

McClelland, McNaughton & O'Reilly (1995); Lee & Wilson (2002); Spens & Burgess (2024)

The missing piece: where does replay come from?

Most computational models build replay in by design.

IN PREVIOUS MODELS

Replay is built in → **Replay trains the cortex** → **Learning improves**

The modeller decides that replay happens and what it does (e.g., Spens & Burgess, 2024).

OUR RESEARCH QUESTION

Fast + slow networks learn a task → **Night: no input, no training signal** → **Replay emerges?**

Nothing tells the networks to replay: does it appear because it helps learning?

Can replay emerge spontaneously from learning?

McClelland, McNaughton & O'Reilly (1995); Spens & Burgess (2024)

NEXT → What could happen at night?

What could happen during the night?

During the day the networks experience A → B → C. Three possible outcomes at night:

1 · VERIDICAL REPLAY

Copies of the day

The fast network repeats the sequences it just experienced, like place-cell replay in rats.

2 · GENERATIVE REPLAY

New but rule-following

It produces new sequences that follow the learned grammar: samples from an internal model.

3 · NO REPLAY

No structure

Night activity carries no clear information; learning relies on the day alone.

Key test: does night activity improve learning the next day?

Wilson & McNaughton (1994); Lee & Wilson (2002); Spens & Burgess (2024)

NEXT → Why this matters

Why this matters

If replay emerges on its own, it is a solution, not an assumption.

FOR NEUROSCIENCE

Why the brain replays

Replay would be what a fast and a slow system naturally do when they must learn a structured world together, explaining why it appears during sleep.

FOR ARTIFICIAL NETWORKS

Learning without forgetting

Networks that learn new things quickly tend to overwrite old knowledge. Offline replay could be a strategy that networks discover for themselves.

Replay may not be an ingredient of learning, but a product of it.

McClelland, McNaughton & O'Reilly (1995); Schapiro et al. (2017)

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The Architecture & The Pipeline

Lead: Alejandro Tabas

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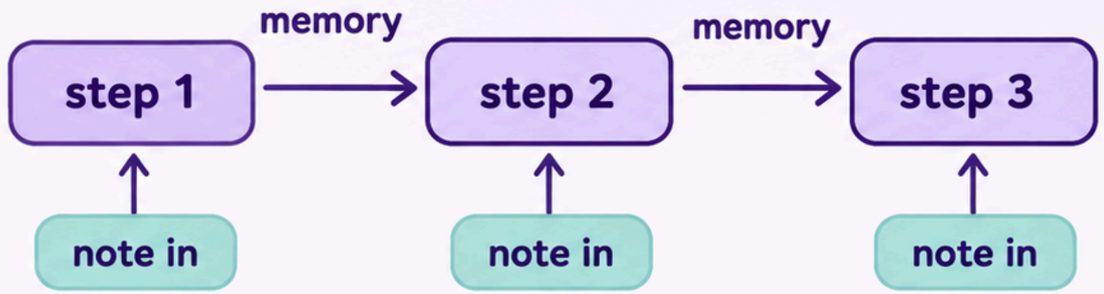
Recurrent Neural Network (RNN)

An RNN is a type of neural network that **remembers** what it has seen before, so it can use that information when processing the next input.

Think of it like reading a sentence:
each word helps you understand the next one!

The same network is used at each time step, and it carries a **hidden state** forward.



Gated Recurrent Unit (GRU)

A **GRU** is a smarter version of an RNN. It uses **"gates"** to decide how much of the past memory to **keep** and how much to **forget**, helping it handle long sequences better.

In short: a GRU is an improved memory unit for **longer context**.

THE CIRCUIT

Three networks (GRUs)

Perception

Sees the noisy input and recovers the clean signal.

Fastest → reacts each moment

Hippocampus

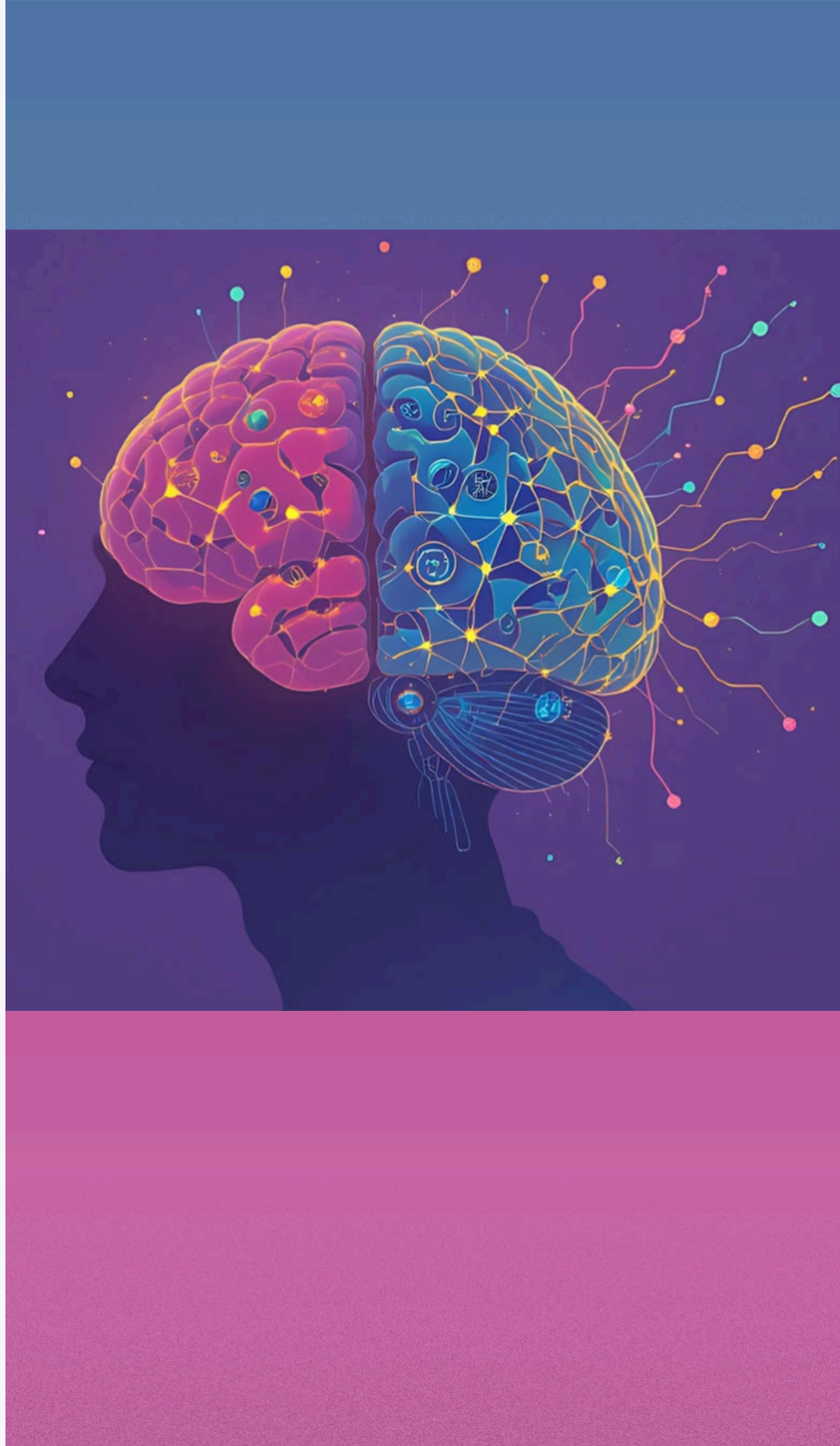
A fast scratchpad memory with limited capacity.

Fast → changes quickly

Cortex / PFC

A large, stable store that builds long-term knowledge.

Slowest → drifts



WHAT THE ARCHITECT TEAM OWNS

Two things to build

nightfever.py

1 · The circuit

Three RNNs; perception, hippocampus and cortex.
→ wired so they exchange signals step by step as a sequence plays.

pipeline.py

2 · The pipeline

The training system: day/night episodes from a schedule, data from the generator, and the loss the networks learn from.

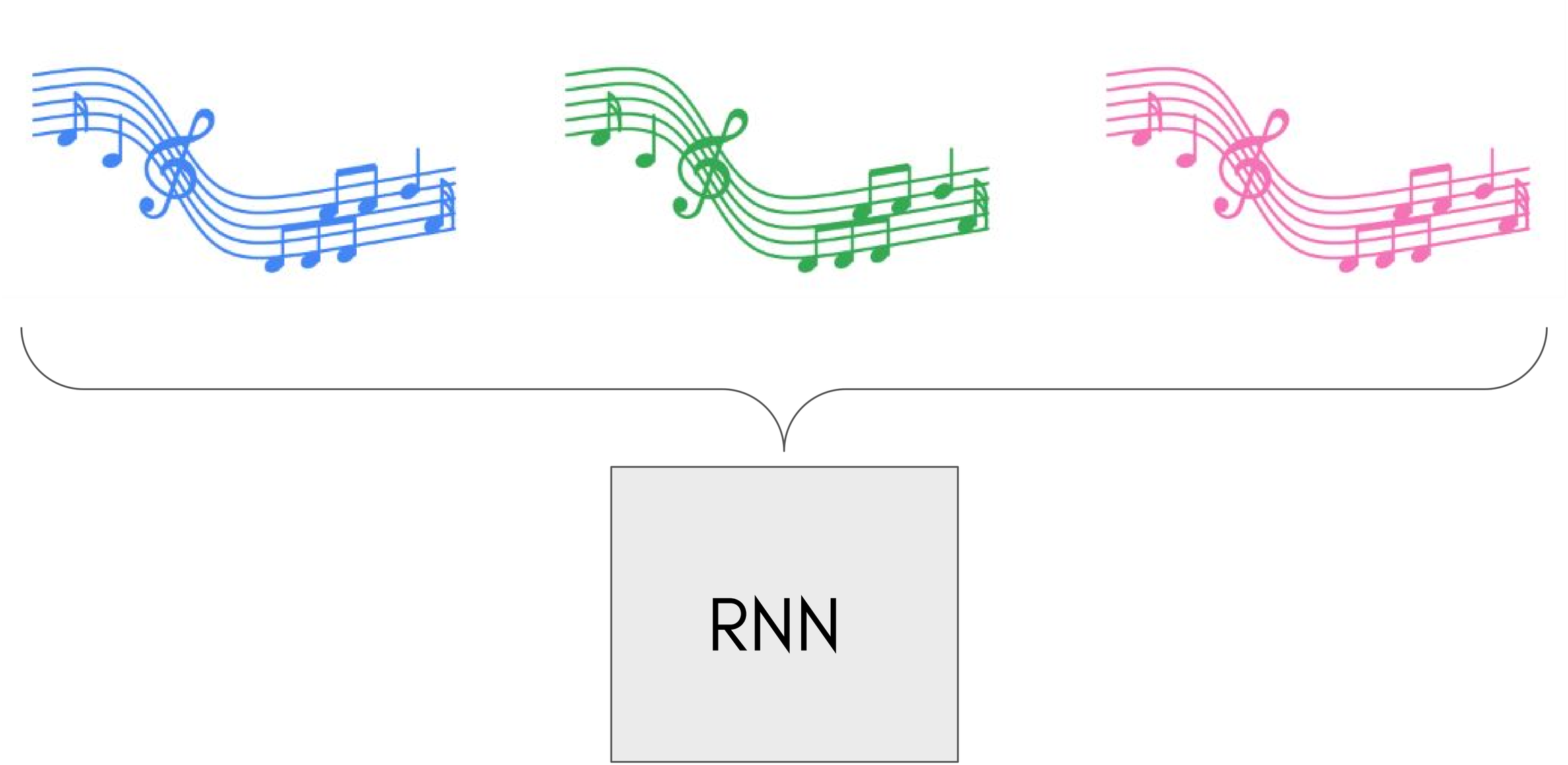
THE PIPELINE

From songwriter to loss

THE PIPELINE

From songwriter to loss

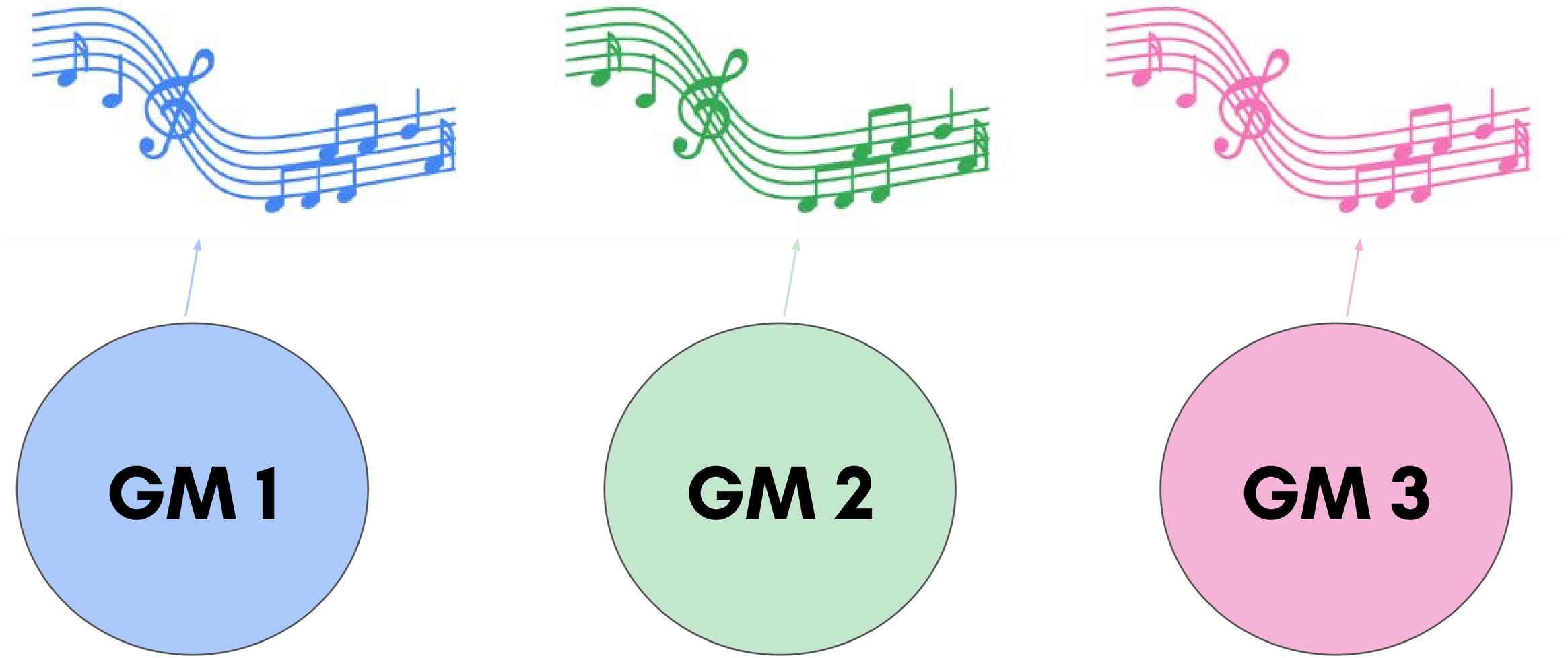
- RNN trains on a series of sequences (songs)



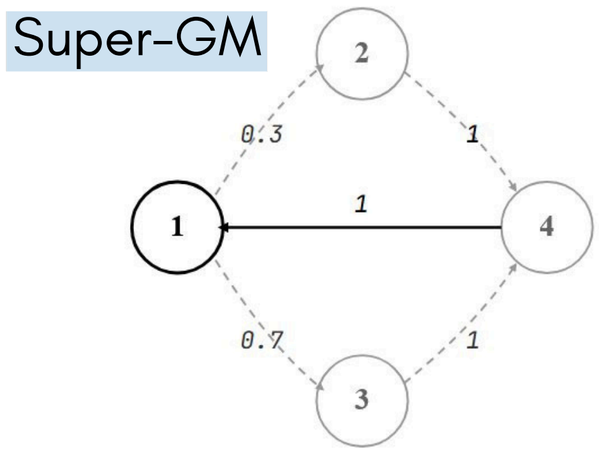
- RNN trains on a series of songs (with noise added!)

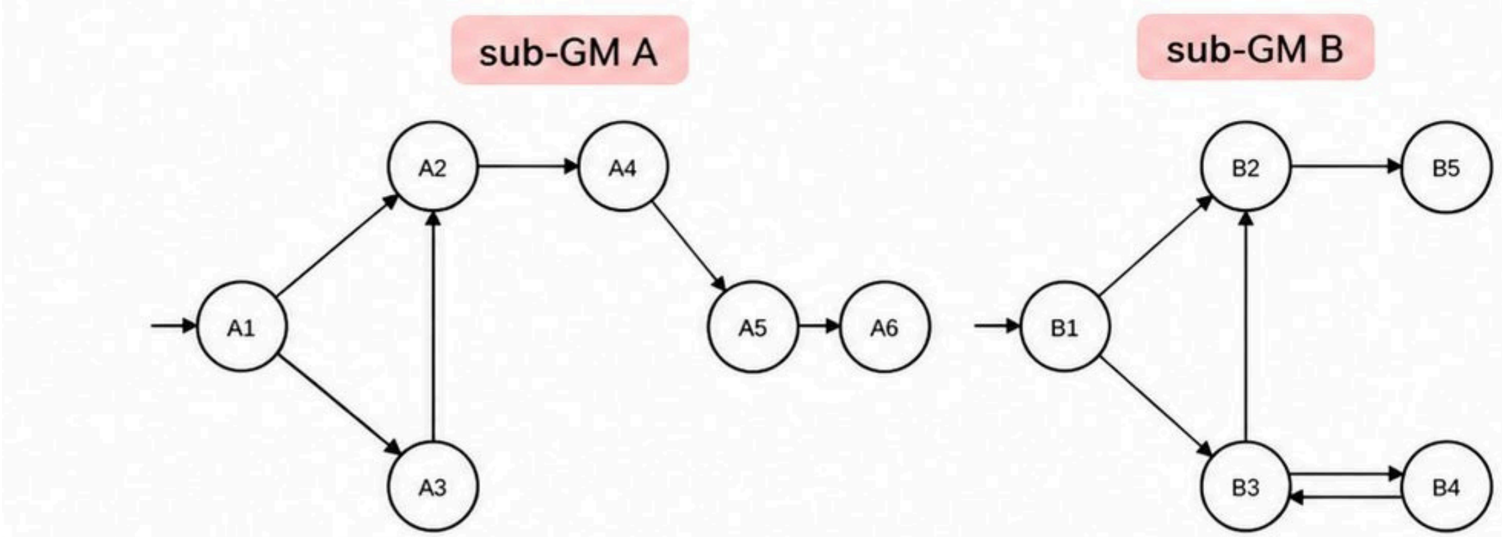
RNN

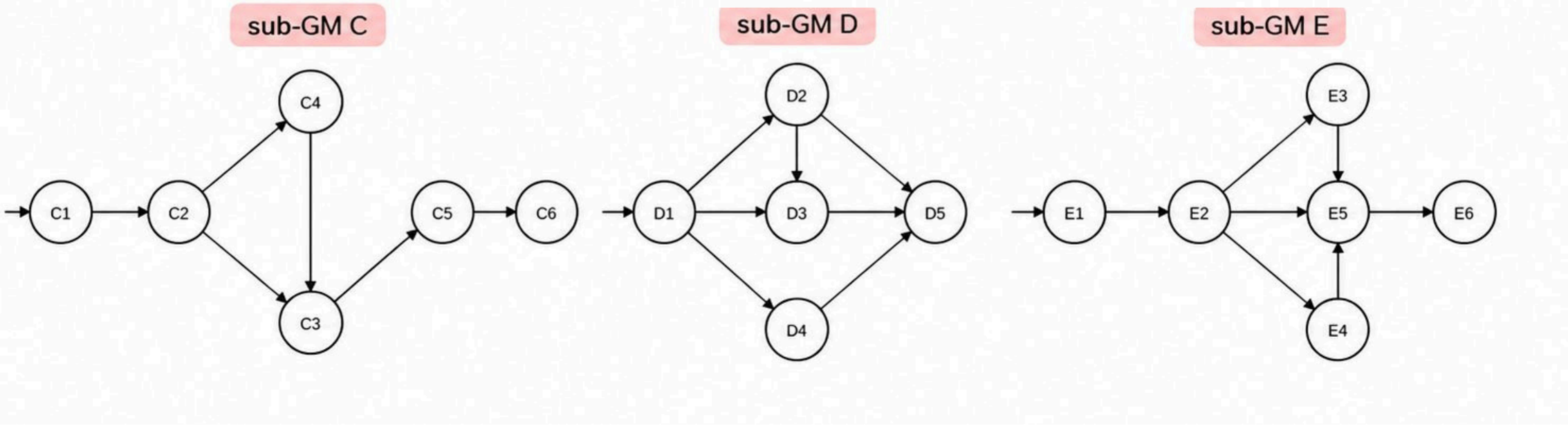
- Each song is sampled from a different generative model (GM)



- Each song is sampled from a super-GM and sub-GMs







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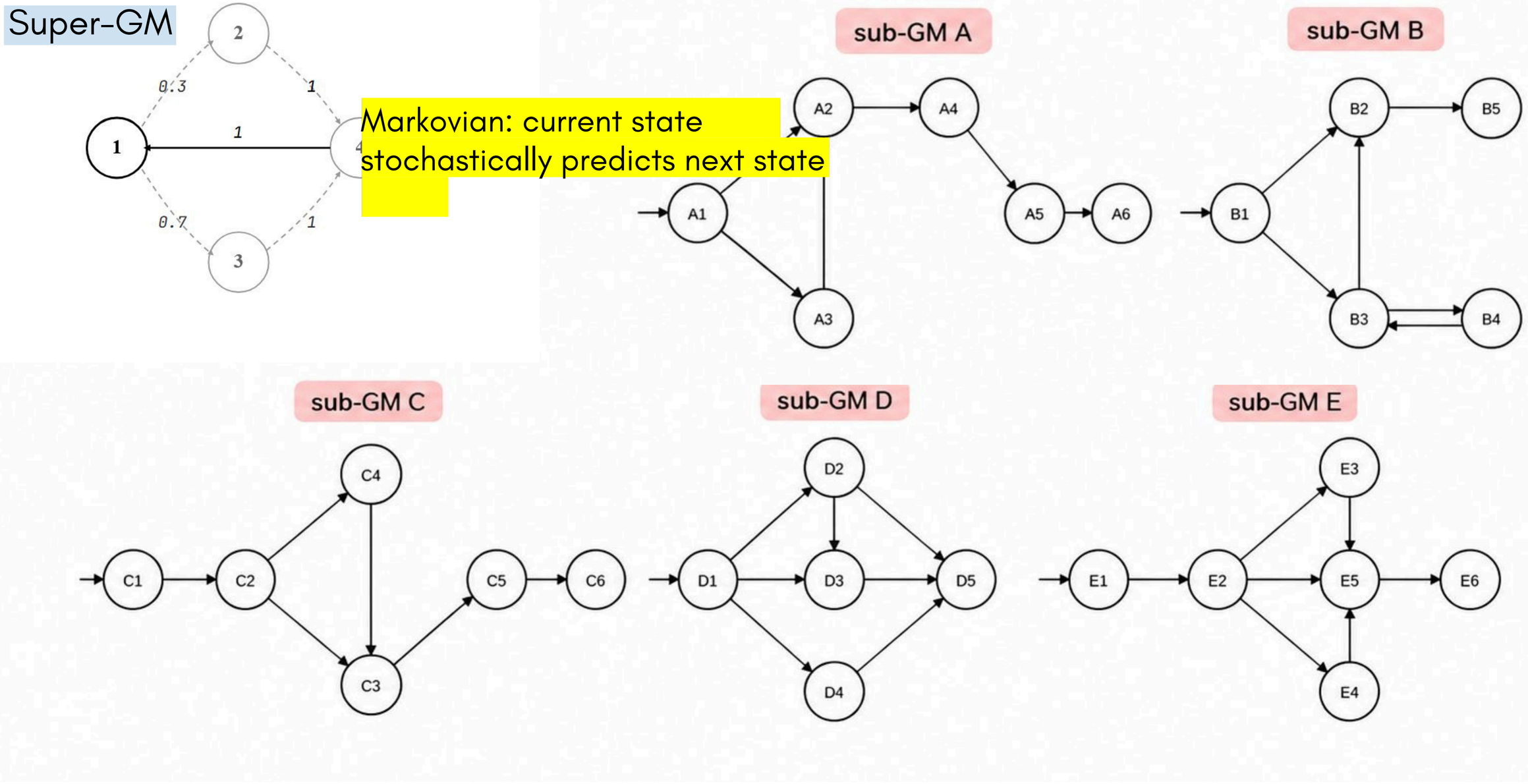
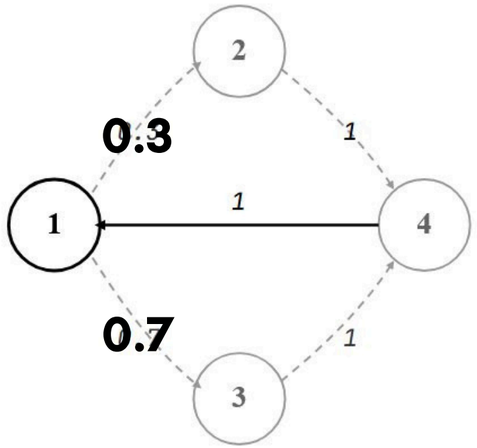
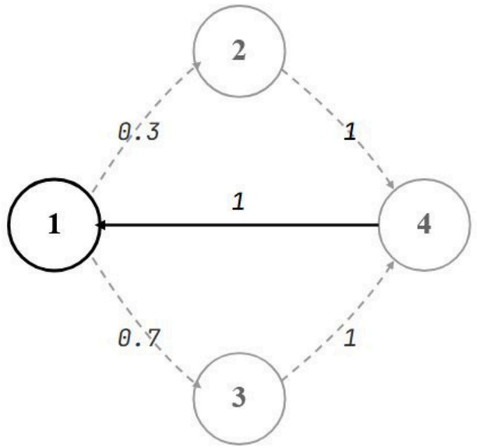


Super-GM



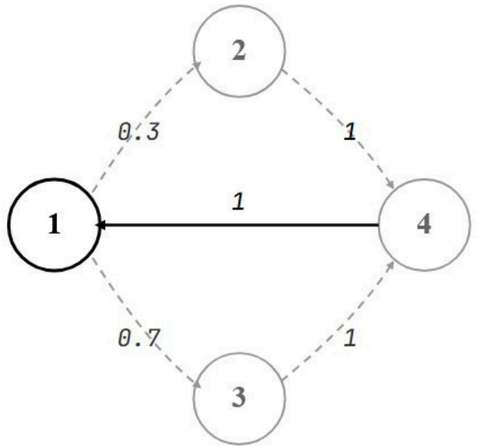
- Sample super sequence with super-GM

Super-GM 1 2 4 1 3 4 1…



- Each Super-GM state is expanded by a sub-GM

Super-GM 1 2 4 1 3 4 1…



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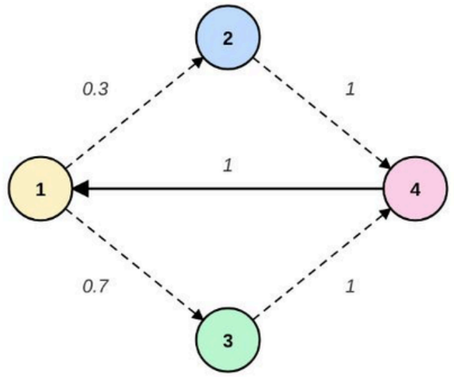

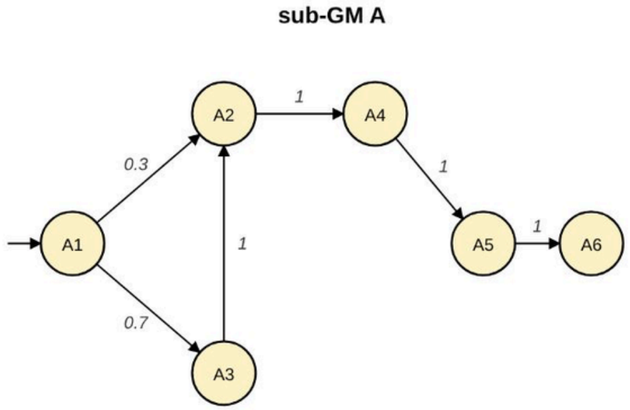

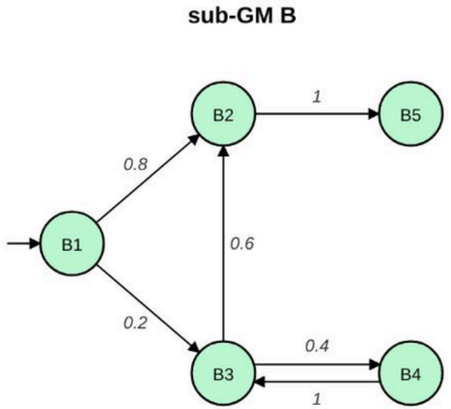

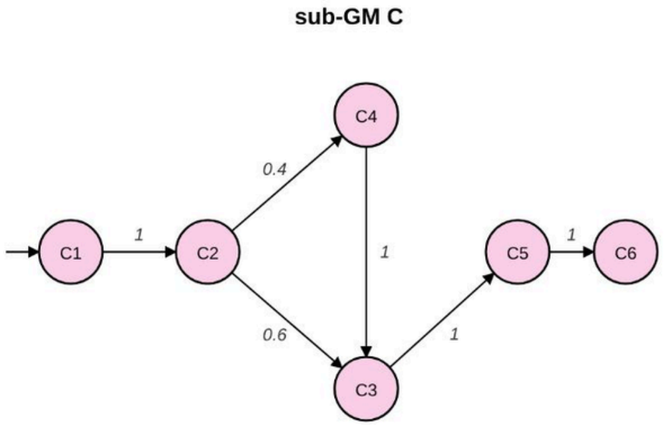

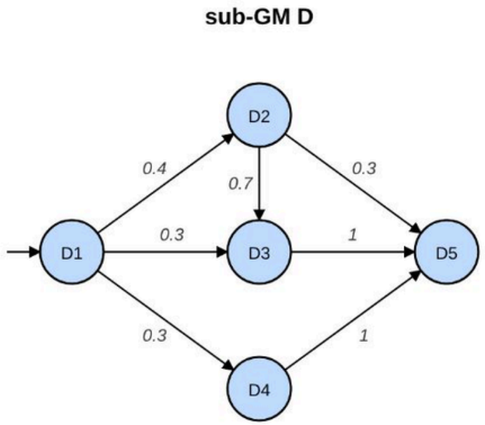

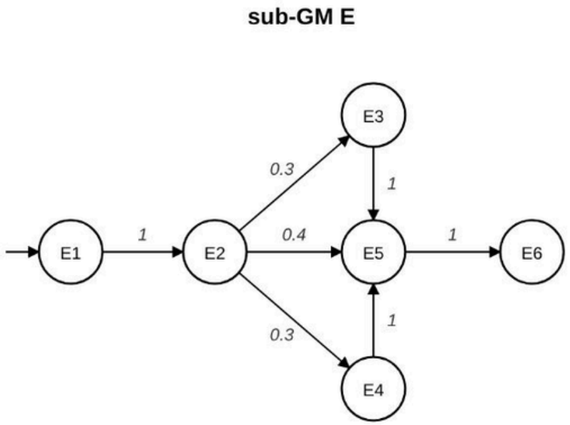

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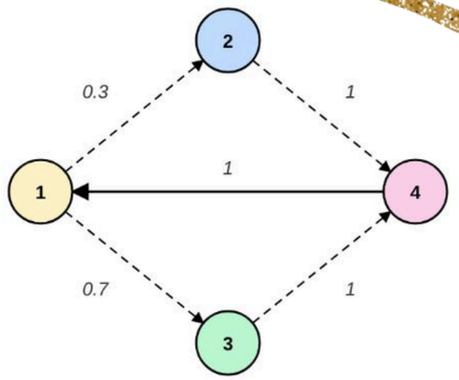

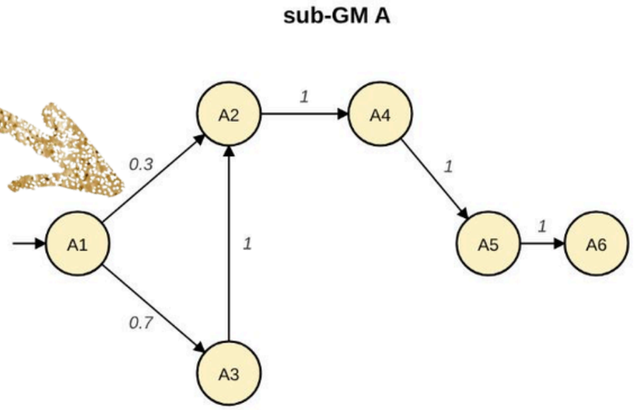

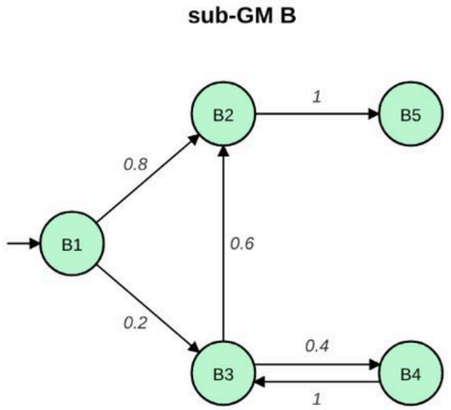

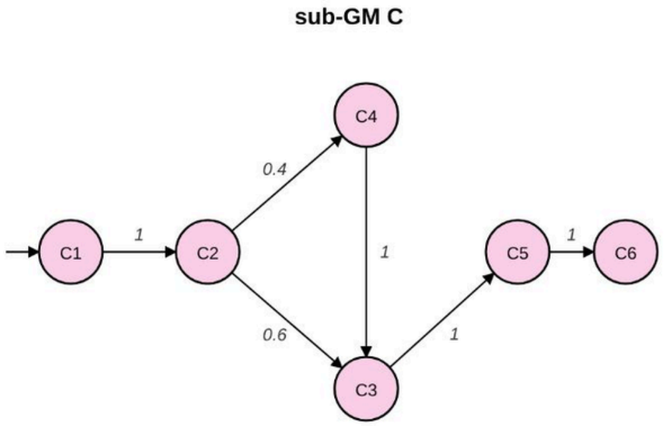

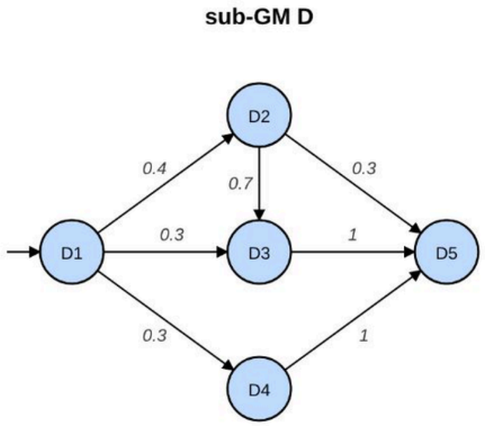

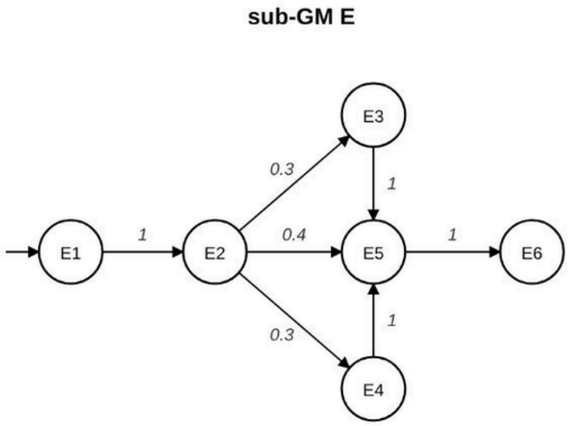

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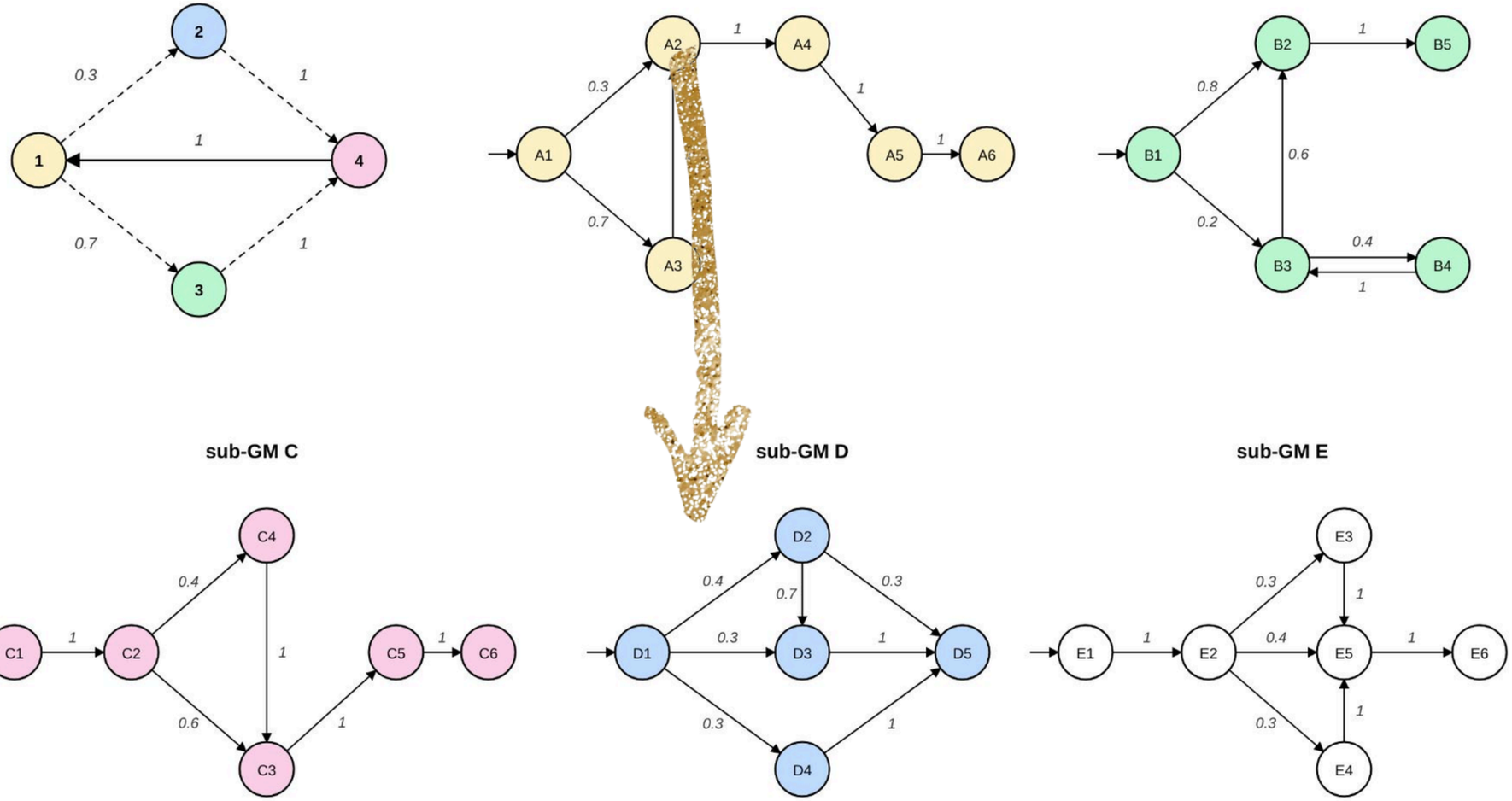
Super-GM A1, A2, A4, A5, A6, 2 4 1 3 4 1…



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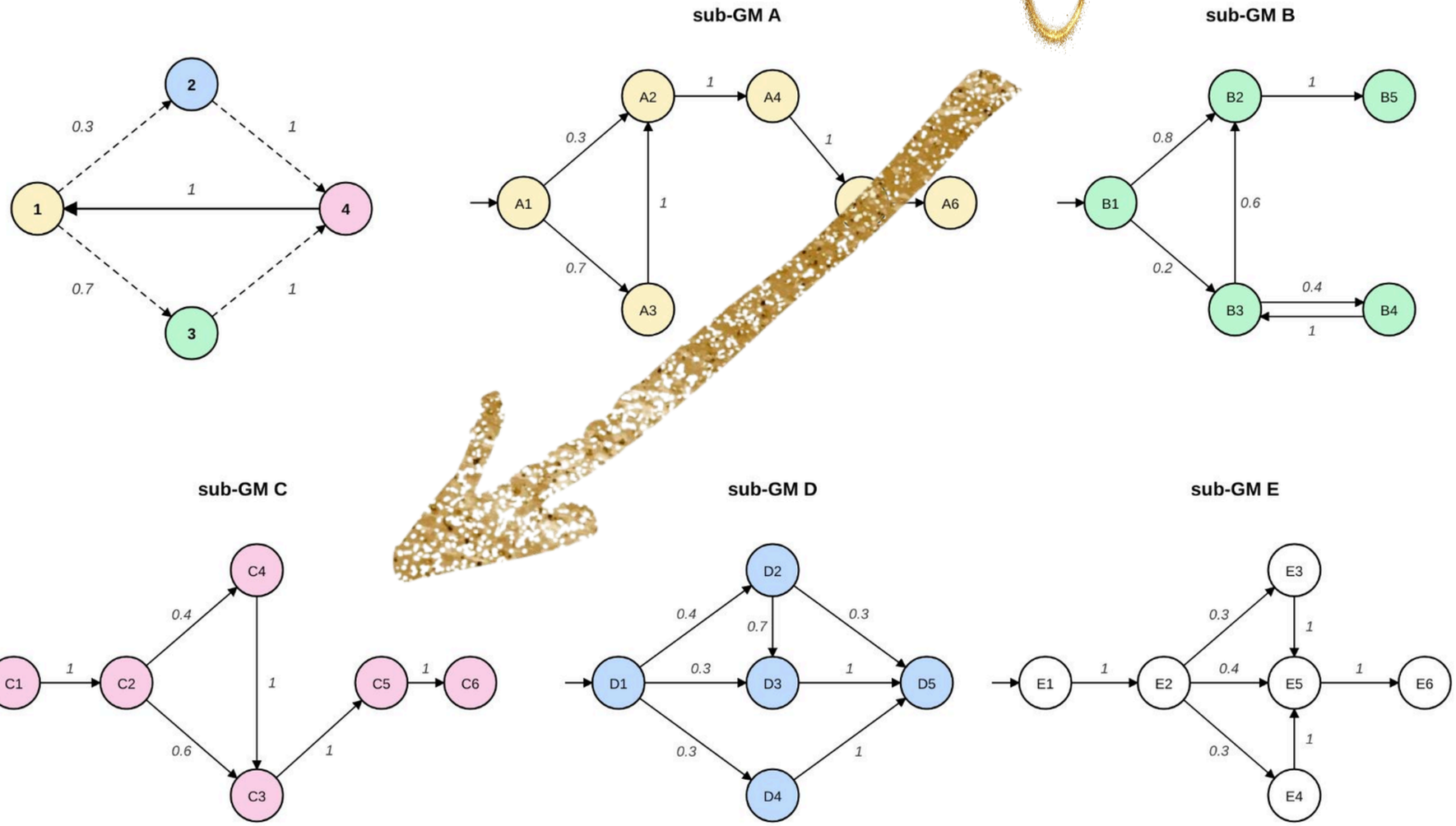
Super-GM A1, A2, A4, A5, A6, D1, D2, D3, D5, 4 1 3 4 1…



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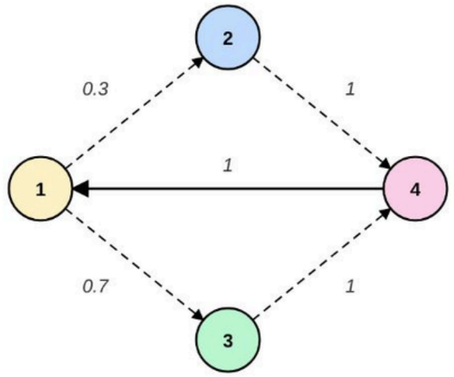
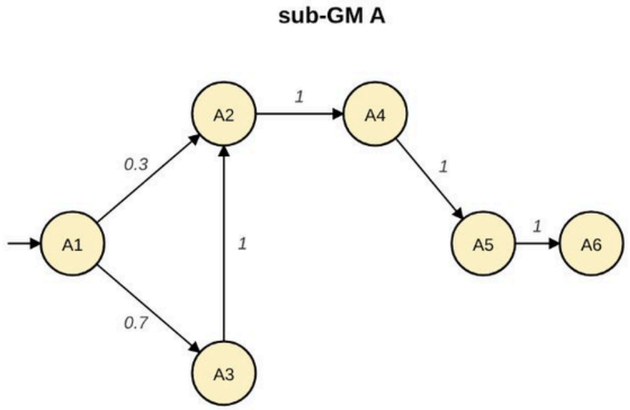
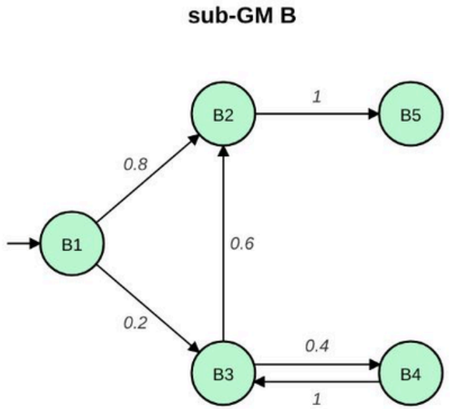
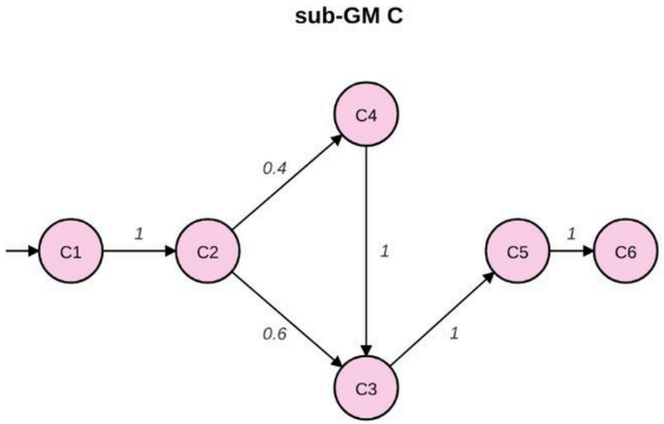
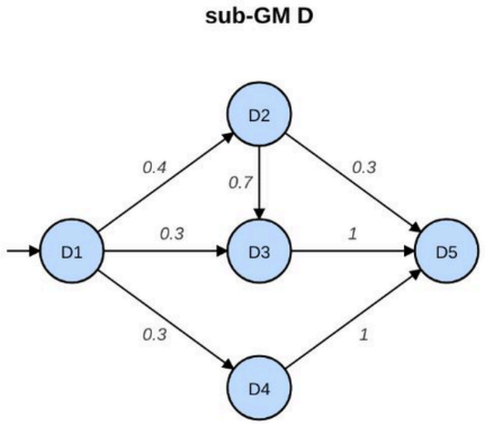
Super-GM A1, A2, A4, A5, A6, D1, D2, D3, D5, C1, C2, C4…, 1 3 4 1…

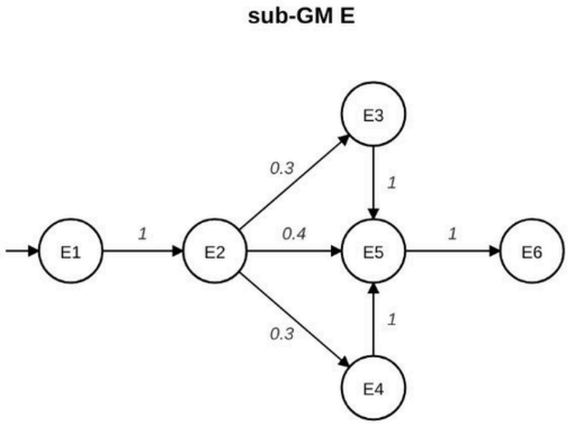












- To make these states readable to the RNN, convert into vectors

A1, A2, A4, A5, A6, D1, D2, D3, D5, C1, C2, C4…

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[0 0 0 1 0 1]

Each sub-GM state maps onto one unique vector

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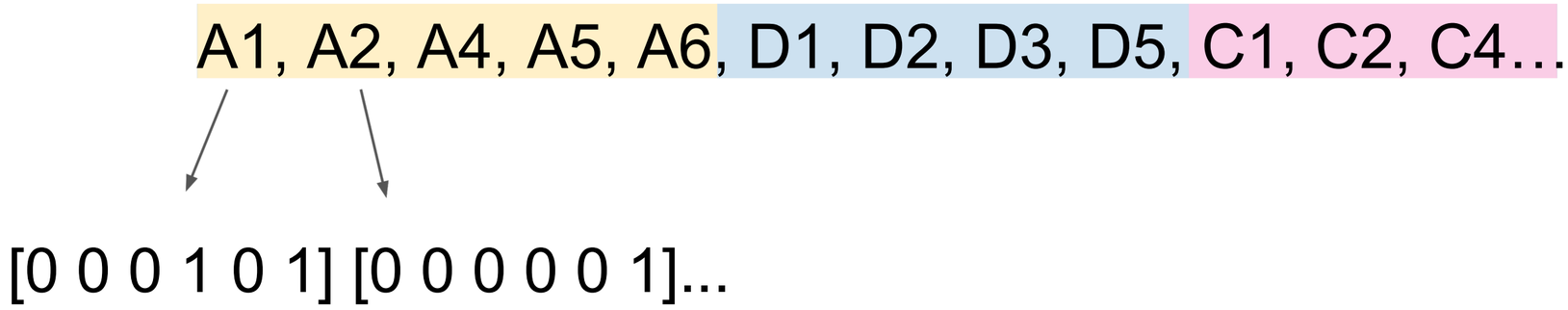

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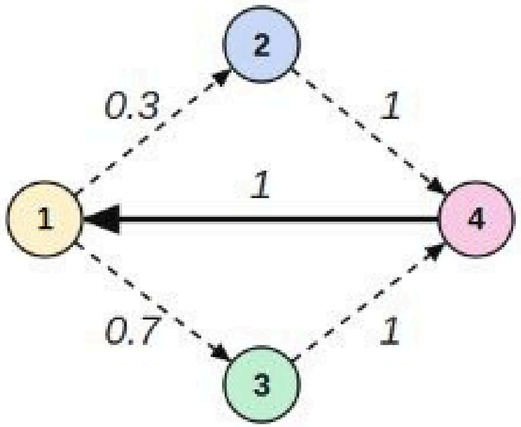
- To make these states readable to the RNN, convert into vectors

A1, A2, A4, A5, A6, D1, D2, D3, D5, C1, C2, C4…

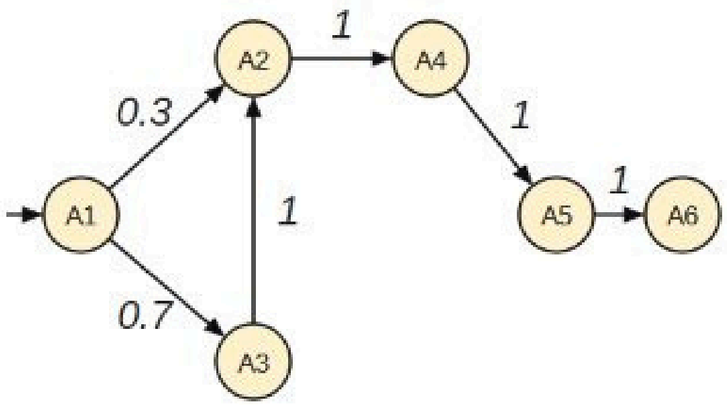
[[0 0 0 1 0 1] [0 0 0 0 0 1] [1 0 0 0 0 1] [0 1 0 1 0 1] [0 1 1 1 1 1]...]

- Super-GM and sub-GM parameters change for each song…

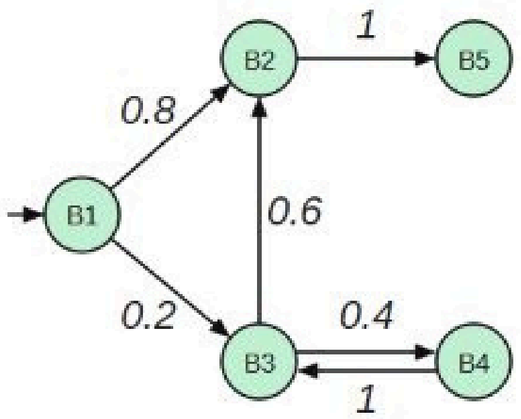
Super-GM



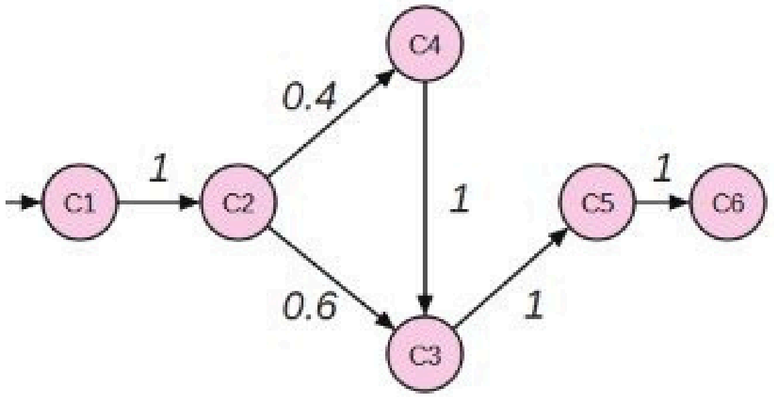
sub-GM A



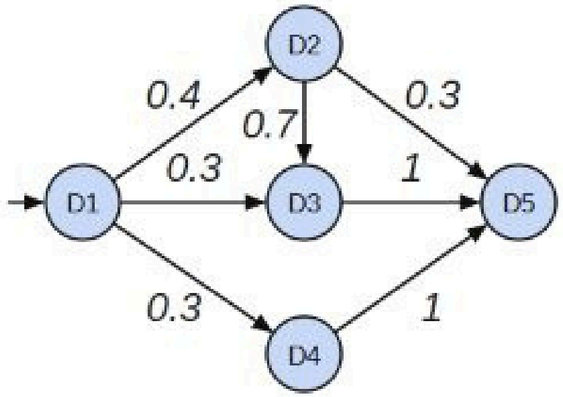
sub-GM B



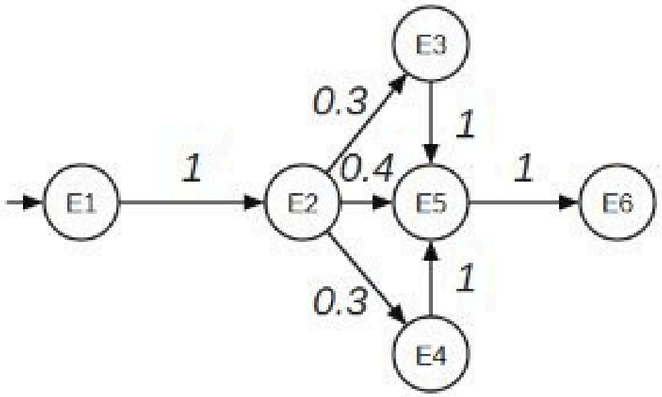
sub-GM C



sub-GM D

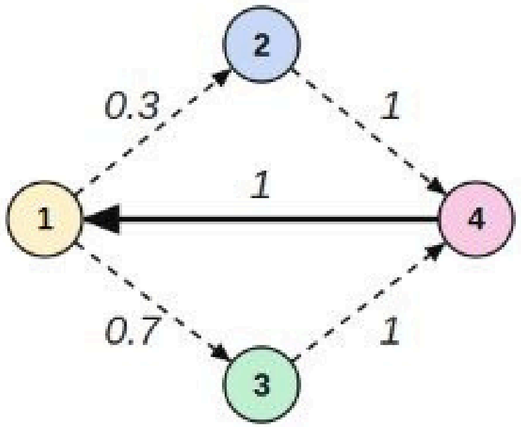


sub-GM E

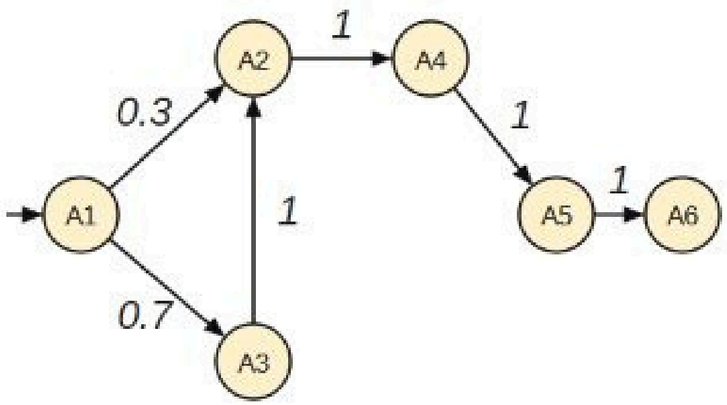


- Super-GM and sub-GM parameters change for each song… how?

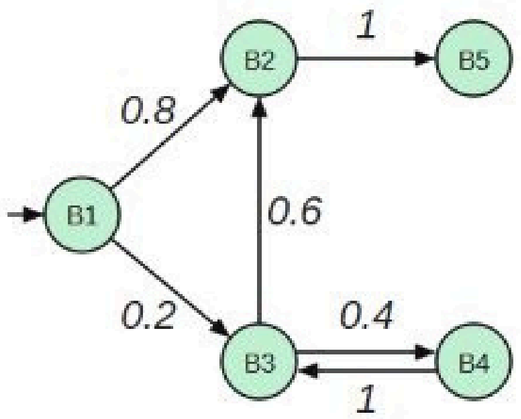
Super-GM



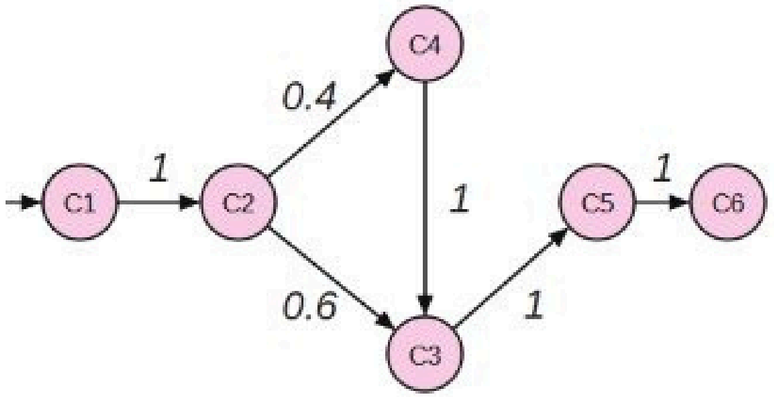
sub-GM A



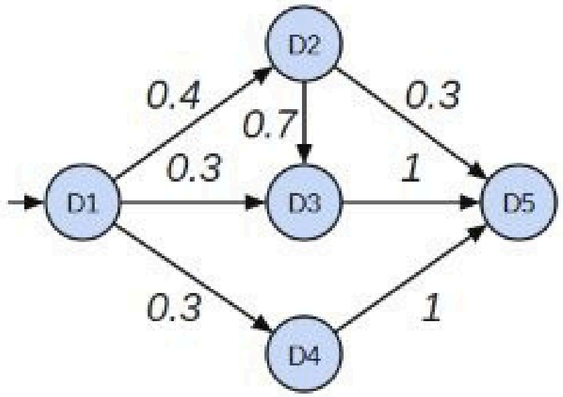
sub-GM B



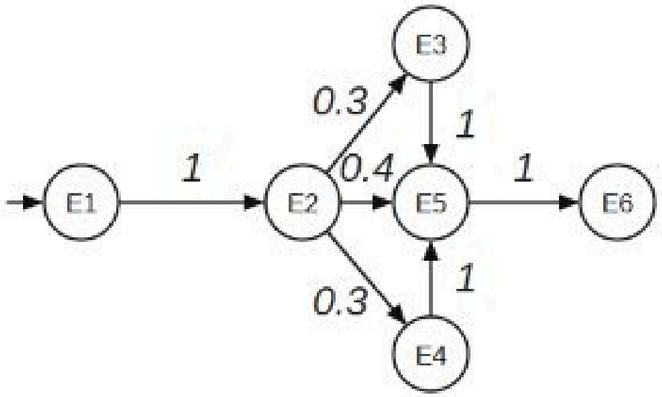
sub-GM C



sub-GM D

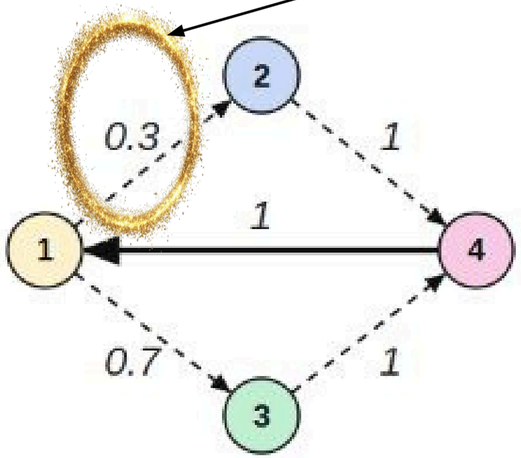


sub-GM E

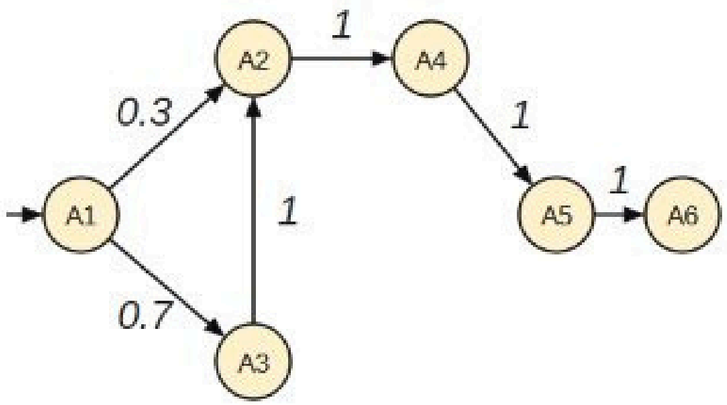


- For each new song, change WEIGHTS of the super-GM and sub-GM

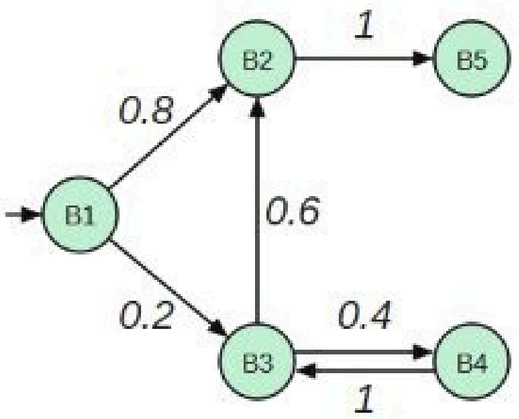
Super-GM



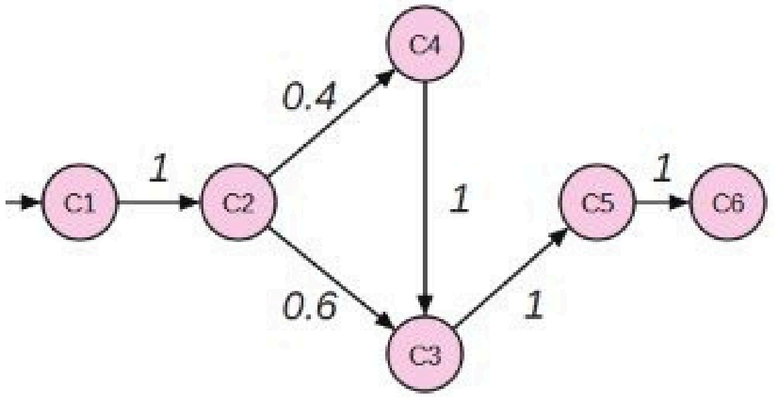
sub-GM A



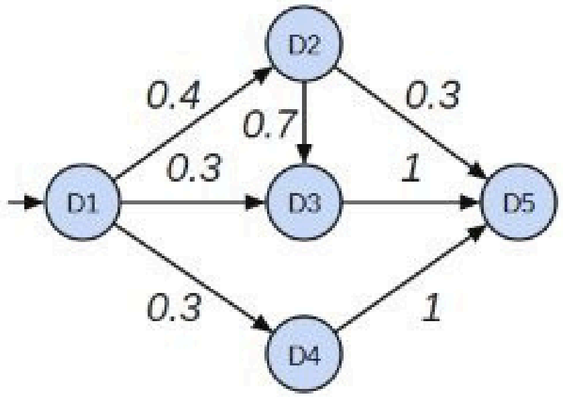
sub-GM B



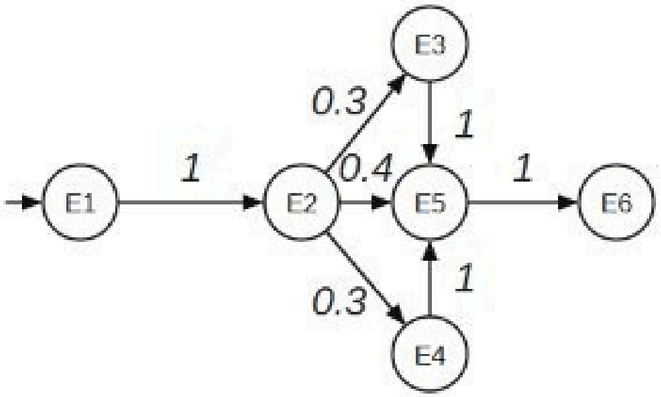
sub-GM C



sub-GM D



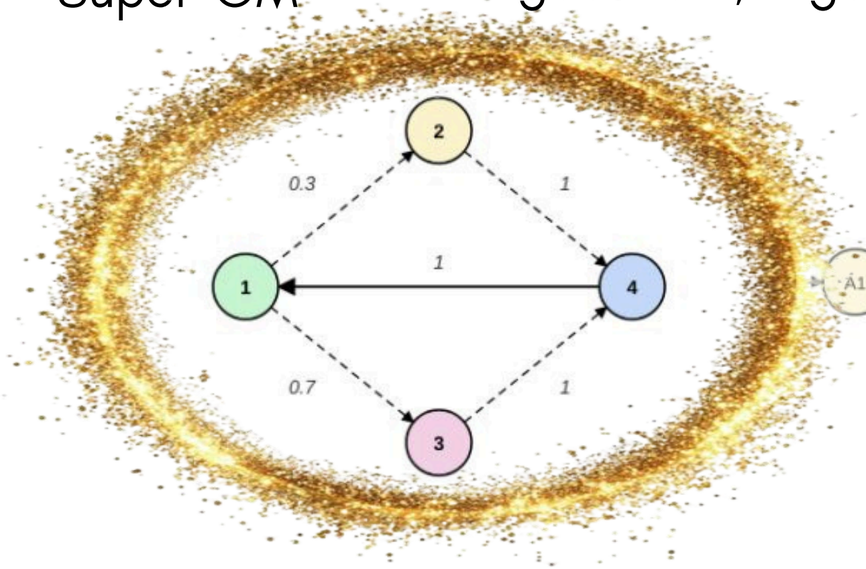
sub-GM E



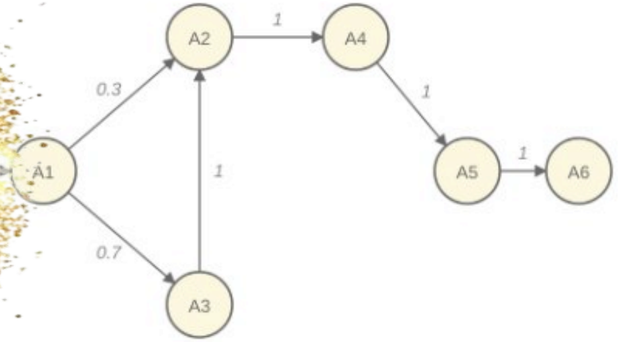
- For each new song, change MAPPINGS between super-GM states and sub-GMs

Super-GM

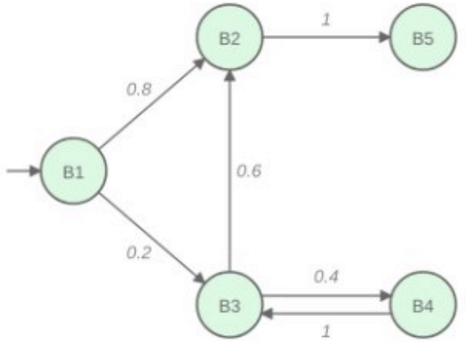
1 goes to B, 2 goes to A, 3 goes to C, 4 goes to D



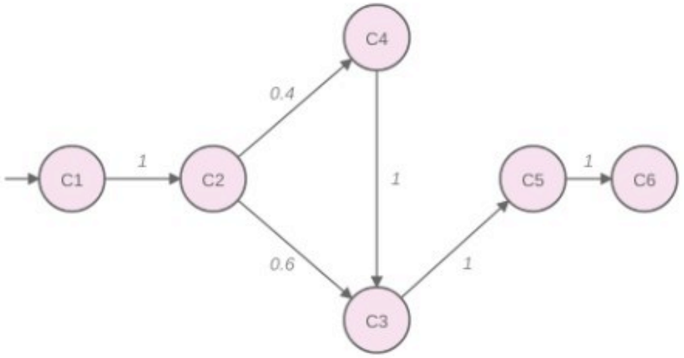
sub-GM A



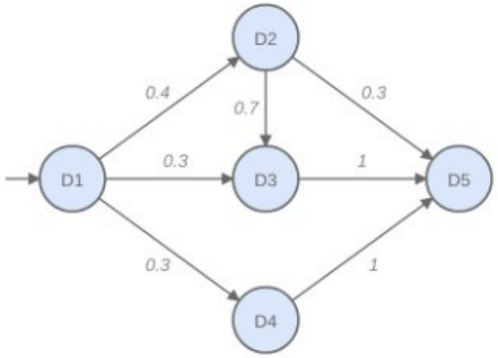
sub-GM B



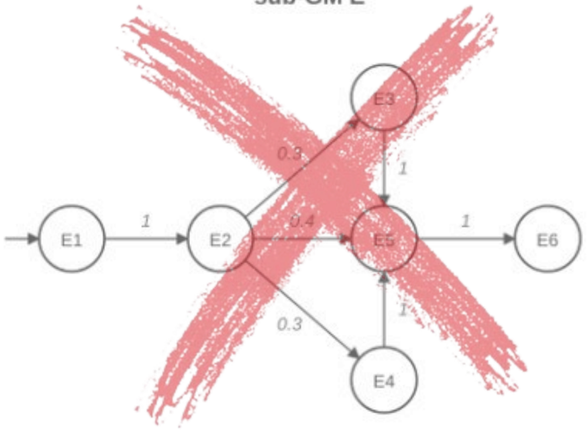
sub-GM C



sub-GM D

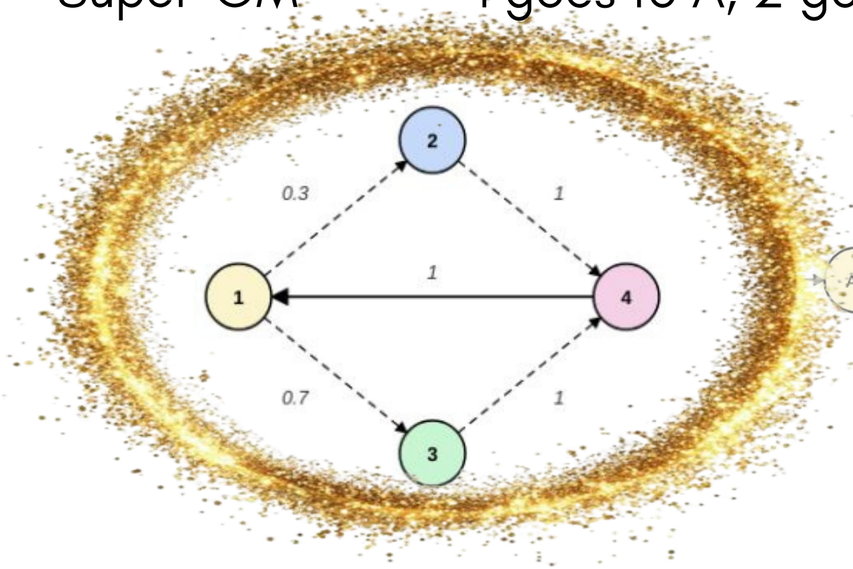


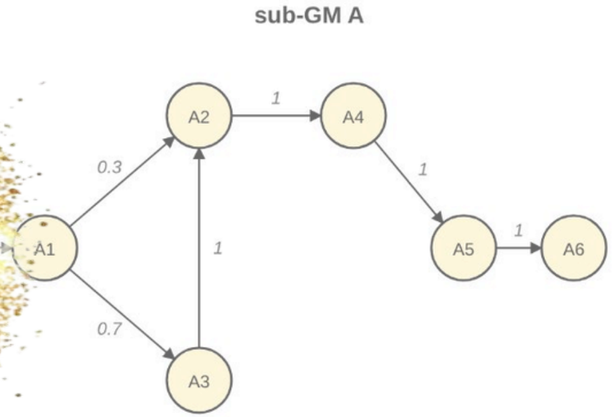
sub-GM E

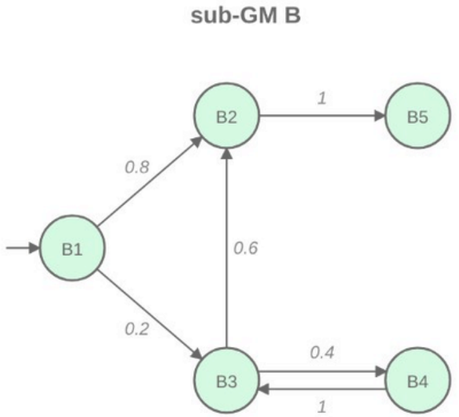


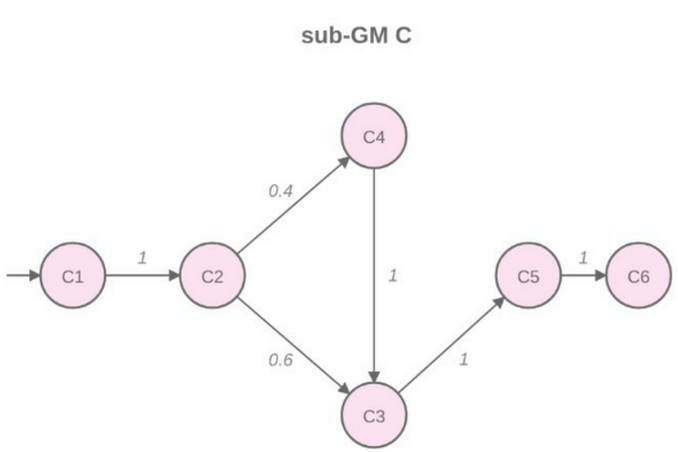
- For each new song, change MAPPINGS between super-GM states and sub-GMs

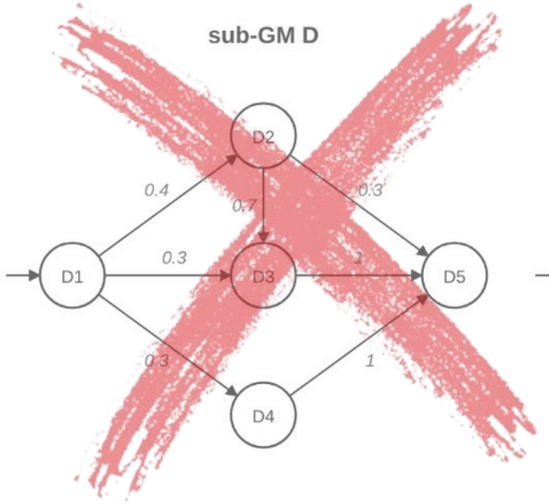
Super-GM 1 goes to A, 2 goes to E, 3 goes to B, 4 goes to C

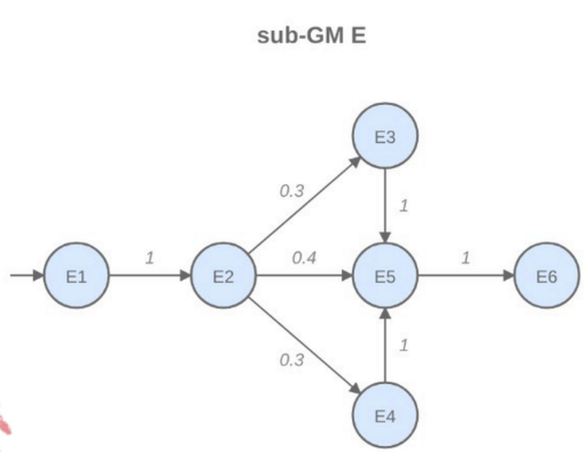


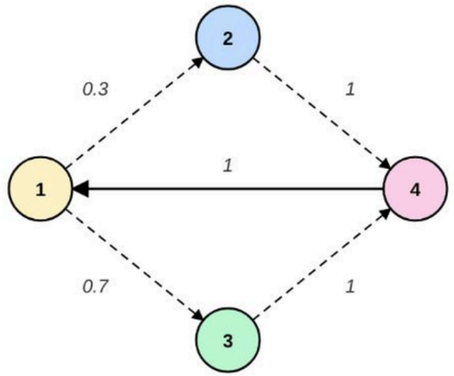




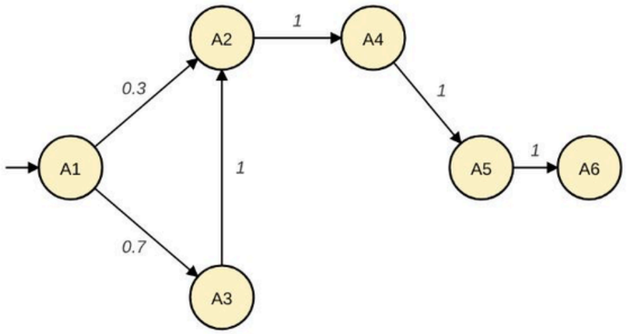




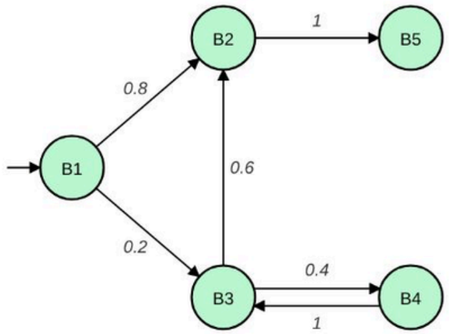




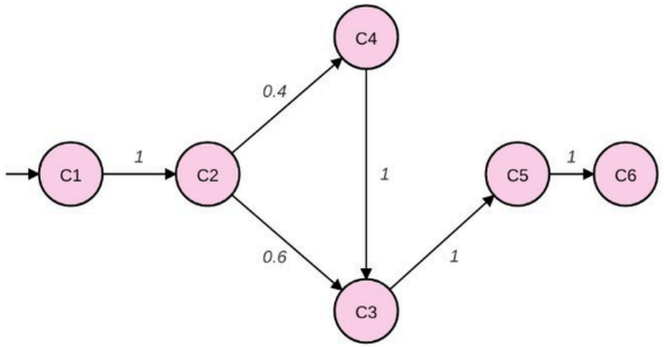
sub-GM A



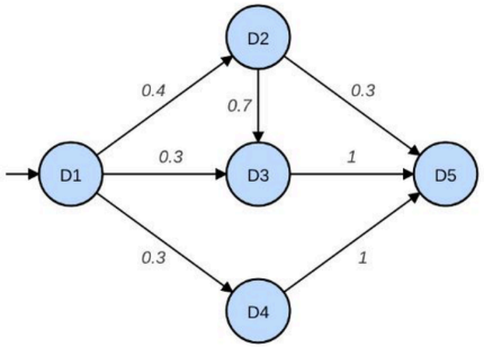
sub-GM B



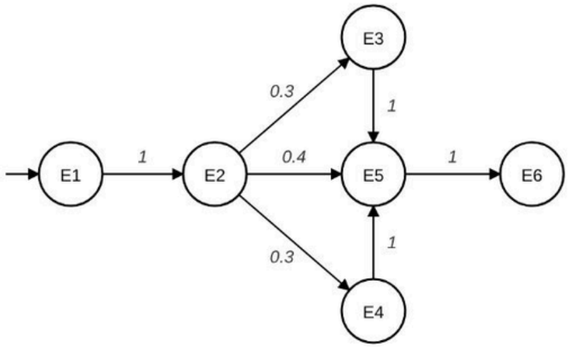
sub-GM C

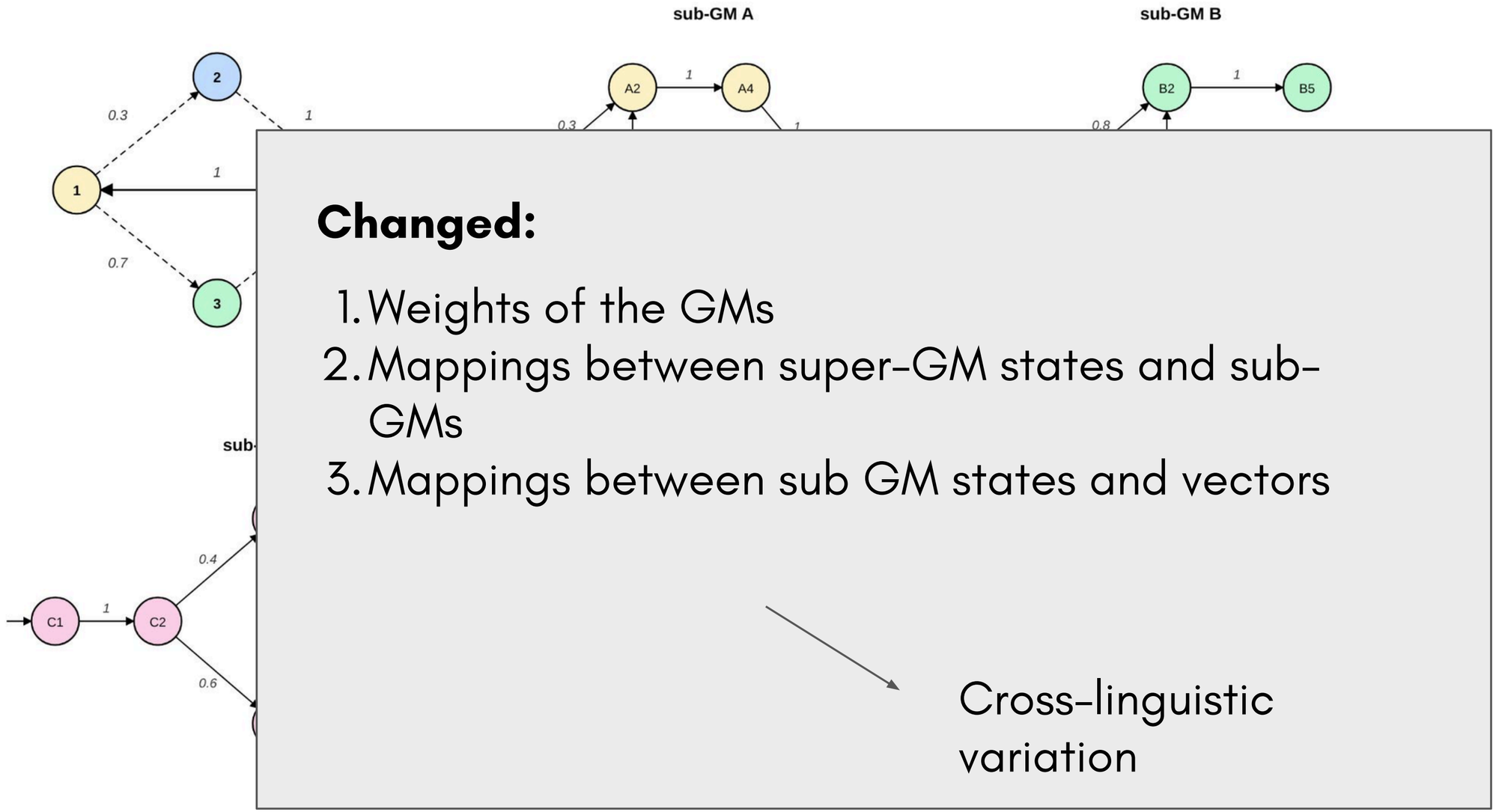


sub-GM D



sub-GM E



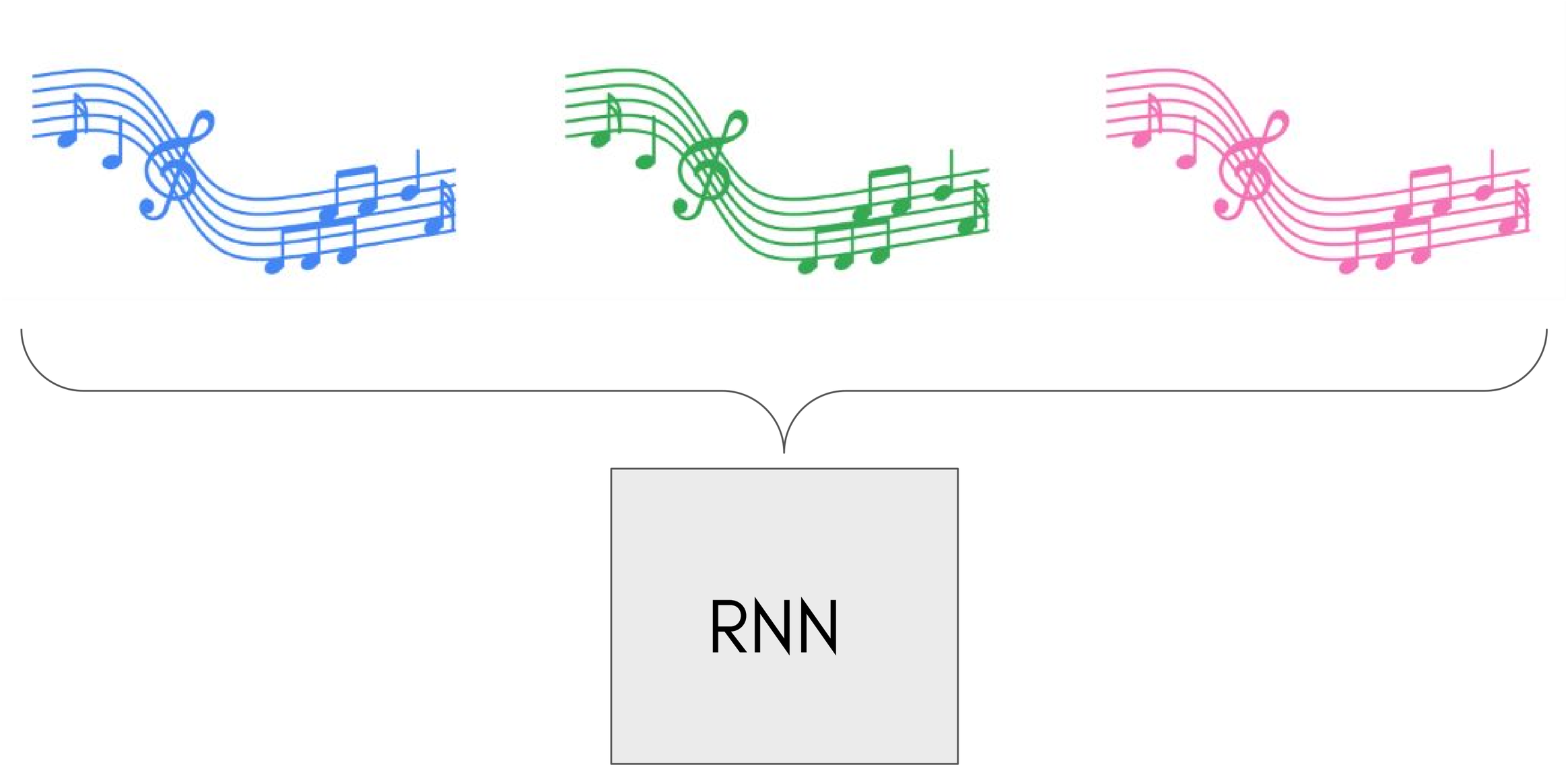


Changed:

1. Weights of the GMs
2. Mappings between super-GM states and sub-GMs
3. Mappings between sub GM states and vectors

Cross-linguistic variation

- RNN trains on a series of songs



- RNN trains on a series of songs (with Gaussian noise added)

RNN

THE PIPELINE

From songwriter to loss

THE PIPELINE

Days and nights, from a schedule

A schedule lists day and night blocks.
One episode: Days carry input and are graded; nights are idle and don’t count.

Day	**Night**	**Day**	**Night**
noisy input · graded	no input · idle	noisy input · graded	no input · idle

Each episode draws fresh worlds from the generator, **so the networks learn to adapt to a world they have never seen;** not to memorize one.

THE LOSS · WHAT WE OPTIMIZE

How Do We Know Our Networks Are Learning?

01 WHAT WE OPTIMIZE

```
loss = how wrong the guess is + a penalty for changing too fast
```

Reconstruction

Reward recovering the clean song from the noisy one.

Slowness

Penalize a network whose state jumps between steps.

BRAINHACK 2026

Results

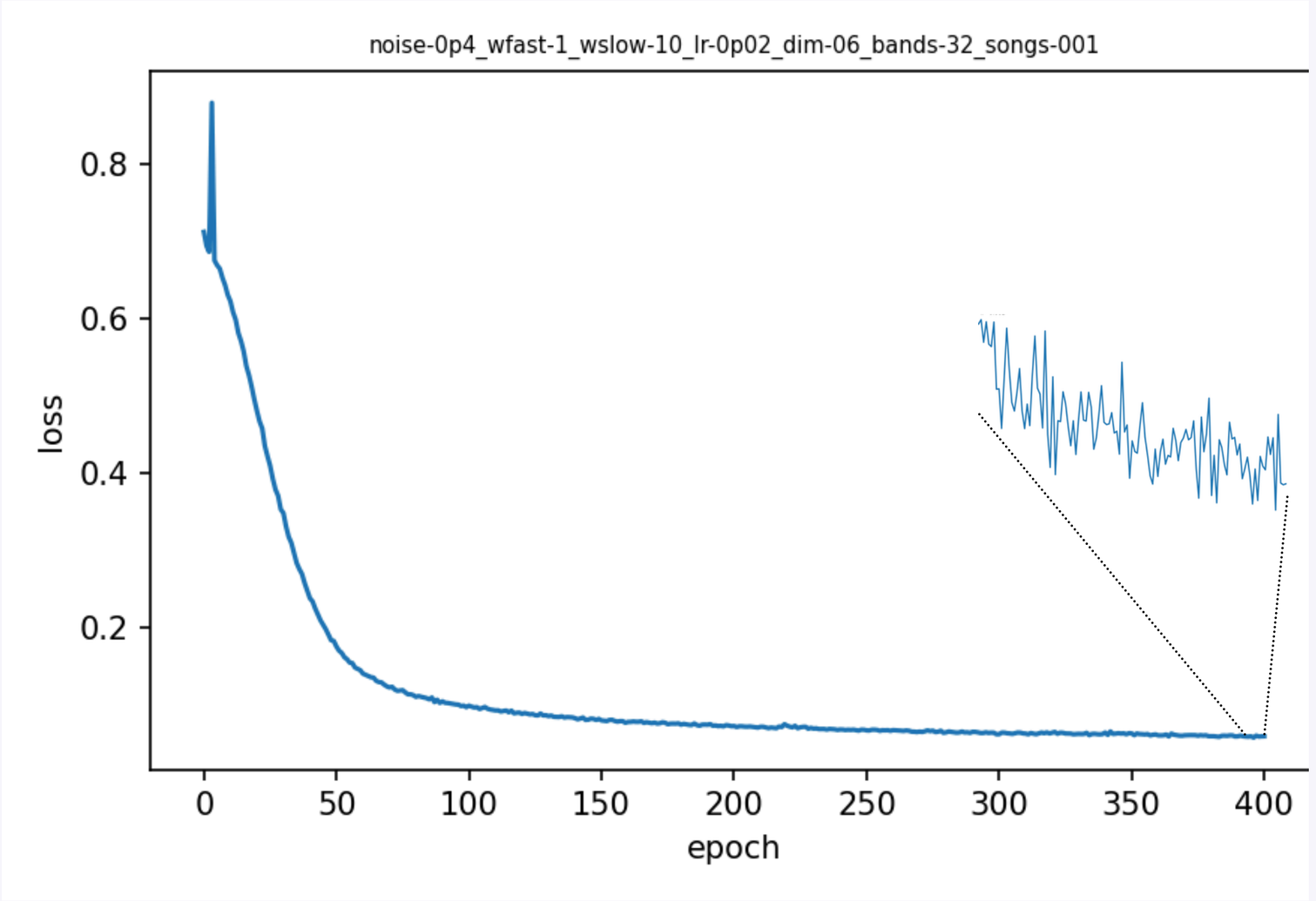
Lead: Alejandro Tabas

Team: Yasin Farshchitabrizi | Outhmane Rassili

THE RESULTS

Figure 1

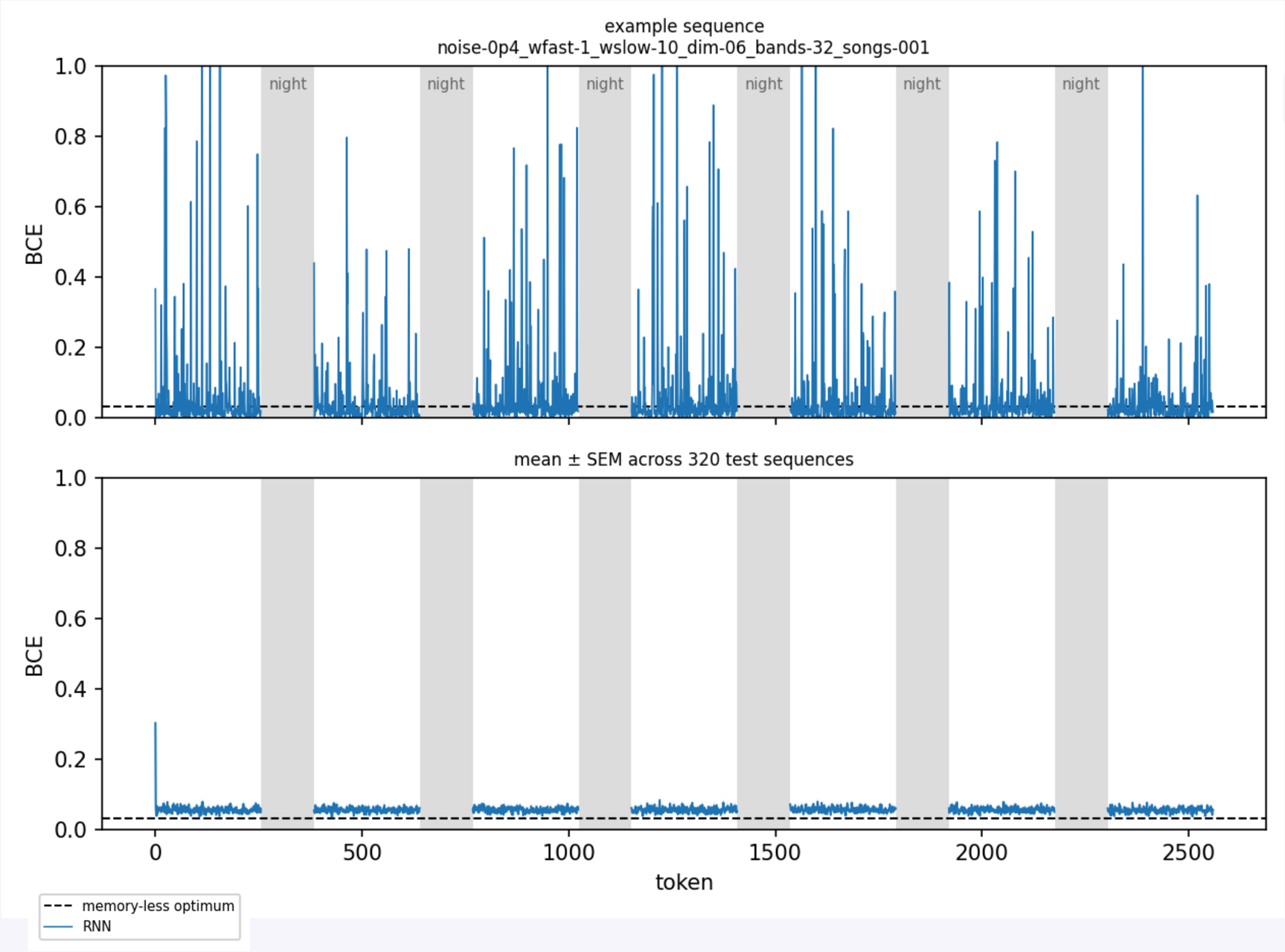
The network is learning...something.



THE RESULTS

Figure 2

But it is not there (YET).



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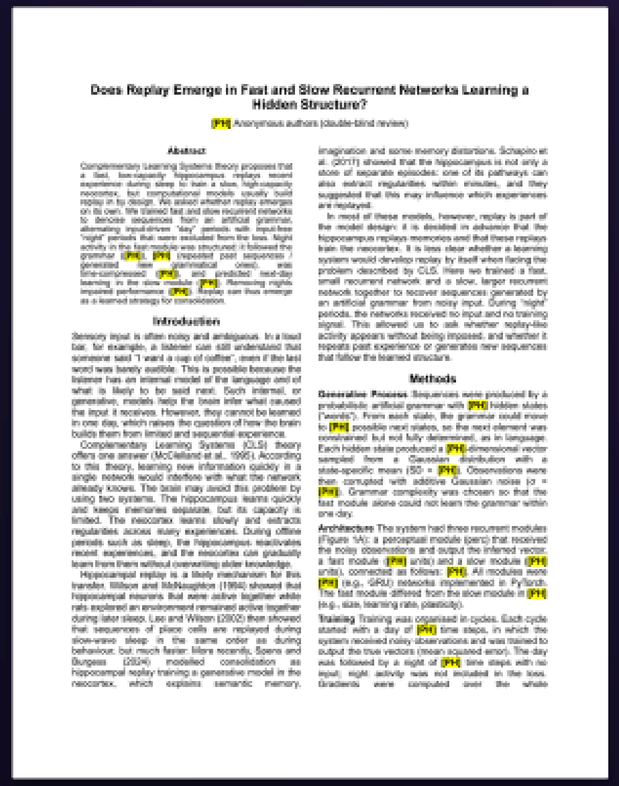
THE RESULTS

Takeaway

- Results are currently ***baking***. Please respect the oven time.
- The team hand-crafted **570 lines** of code (yes, by hand!)
- We successfully trained **our very own RNN circuit**. It's alive!
- All coding infrastructure is officially **up** and running 🏃

We also drafted a CCN extended abstract

Written during the hackathon, ready to be completed with our results.



Does Replay Emerge in Fast and Slow Recurrent Networks Learning a Hidden Structure?

[PH] Anonymous authors (double-blind review)

Abstract

Complementary Learning Systems theory proposes that a fast, low-capacity hippocampus replays recent experience during sleep to train a slow, high-capacity neocortex, but computational models usually build replay in by design. We asked whether replay emerges on its own. We trained fast and slow recurrent networks to denoise sequences from an artificial grammar, alternating input-driven "day" periods with input-free "night" periods that were excluded from the loss. Night activity in the fast module was structured: it followed the grammar ([PH]), [PH] (repeated past sequences / generalized new grammatical ones), was time-compressed ([PH]), and predicted next-day learning in the slow module ([PH]). Removing nights impaired performance ([PH]). Replay can thus emerge as a learned strategy for consolidation.

Introduction

Sensory input is often noisy and ambiguous. In a loud bar, for example, a listener can still understand that someone said "I want a cup of coffee", even if the last word was barely audible. This is possible because the listener has an internal model of the language and of what is likely to be said next. Such internal, or generative, models help the brain infer what caused the input it receives. However, they cannot be learned in one day, which raises the question of how the brain builds them from limited and sequential experience.

Complementary Learning Systems (CLS) theory offers one answer (McClelland et al., 1995). According to this theory, learning new information quickly in a single network would interfere with what the network already knows. The brain may avoid this problem by using two systems. The hippocampus learns quickly and keeps memories separate, but its capacity is limited. The neocortex learns slowly and extracts regularities across many experiences. During offline periods such as sleep, the hippocampus reactivates recent experiences, and the neocortex can gradually learn from them without overwriting older knowledge.

Hippocampal replay is a likely mechanism for this. Ikenden Wilson and McNaughton (1994) showed that hippocampal neurons that were active together while rats explored an environment remained active together during later sleep. Lee and Wilson (2002) then showed that sequences of place cells are replayed during slow-wave sleep in the same order as during behaviour, but much faster. More recently, Spens and Burgess (2024) modelled consolidation as hippocampal replay training a generative model in the neocortex, which explains semantic memory, imagination and some memory distortions. Schapiro et al. (2017) showed that the hippocampus is not only a store of separate episodes: one of its pathways can also extract regularities within minutes, and they suggested that this may influence which experiences are replayed.

In most of these models, however, replay is part of the model design: it is decided in advance that the hippocampus replays memories and that these replays train the neocortex. It is less clear whether a learning system would develop replay by itself when facing the problem described by CLS. Here we trained a fast, small recurrent network and a slow, larger recurrent network together to recover sequences generated by an artificial grammar from noisy input. During "night" periods, the networks received no input and no training signal. This allowed us to ask whether replay-like activity appears without being imposed, and whether it repeats past experience or generates new sequences that follow the learned structure.

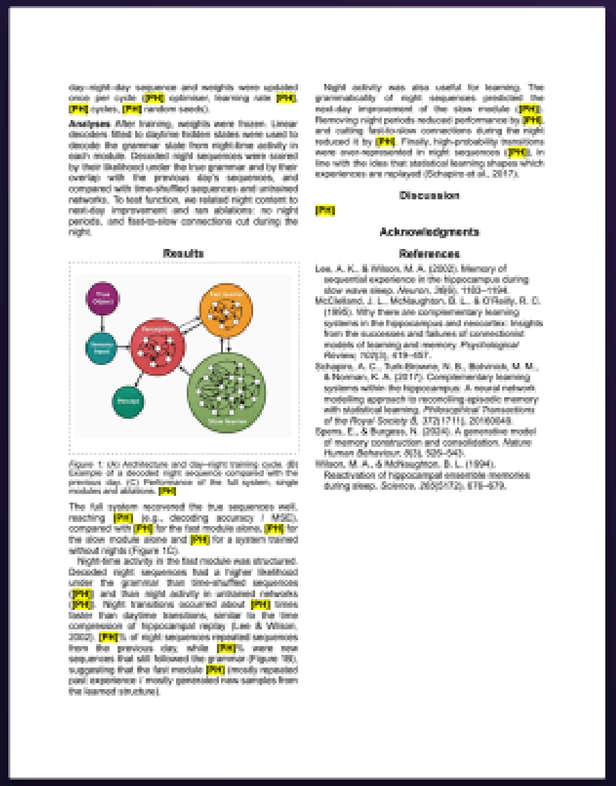
Methods

Generative Process Sequences were produced by a probabilistic artificial grammar with [PH] hidden states ("events"). From each state, the grammar could move to [PH] possible next states, so the next element was constrained but not fully determined, as in language. Each hidden state produced a [PH]-dimensional vector sampled from a Gaussian distribution with a state-specific mean (SD = [PH]). Observations were then corrupted with additive Gaussian noise (σ = [PH]). Grammar complexity was chosen so that the fast module alone could not learn the grammar within one day.

Architecture The system had three recurrent modules (Figure 1A): a perceptual module (perc) that received the noisy observations and output the inferred vector, a fast module ([PH] units) and a slow module ([PH] units), connected as follows: [PH]. All modules were [PH] (e.g., GRU) networks implemented in PyTorch. The fast module differed from the slow module in [PH] (e.g., size, learning rate, plasticity).

Training Training was organised in cycles. Each cycle started with a day of [PH] time steps, in which the system received noisy observations and was trained to output the true vectors (mean squared error). The day was followed by a night of [PH] time steps with no input; night activity was not included in the loss. Gradients were computed over the whole

Page 1



day–night–day sequence and weights were updated once per cycle ([PH] optimiser, learning rate [PH], [PH] cycles, [PH] random seeds).

Analyses After training, weights were frozen. Linear decoders fitted to daytime hidden states were used to decode the grammar state from night-time activity in each module. Decoded night sequences were scored by their likelihood under the true grammar and by their overlap with the previous day's sequences, and compared with time-shuffled sequences and untrained networks. To test function, we retrained night content to next-day improvement and ran ablations: no night periods, and fast-to-slow connections cut during the night.

Results

Figure 1: (A) Architecture and day–night training cycle. (B) Example of a decoded night sequence compared with the previous day. (C) Performance of the full system, single modules and ablations. [PH]

The full system recovered the true sequences well, reaching [PH] (e.g., decoding accuracy / MSE), compared with [PH] for the fast module alone, [PH] for the slow module alone and [PH] for a system trained without nights (Figure 1C).

Night-time activity in the fast module was structured. Decoded night sequences had a higher likelihood under the grammar than time-shuffled sequences ([PH]) and than night activity in untrained networks ([PH]). Night transitions occurred about [PH] times faster than daytime transitions, similar to the time compression of hippocampal replay (Lee & Wilson, 2002). [PH]% of night sequences reinstated sequences from the previous day, while [PH]% were new sequences that still followed the grammar (Figure 1B), suggesting that the fast module [PH] (mostly repeated past experience / mostly generalized new samples from the learned structure).

Night activity was also useful for learning. The grammaticality of night sequences predicted the next-day improvement of the slow module ([PH]). Removing night periods reduced performance by [PH], and cutting fast-to-slow connections during the night reduced it by [PH]. Finally, high-probability transitions were over-represented in night sequences ([PH]), in line with the idea that statistical learning shapes which experiences are replayed (Schapiro et al., 2017).

Discussion

[PH]

Acknowledgments

References

Lee, A. K., & Wilson, M. A. (2002). Memory of sequential experience in the hippocampus during slow wave sleep. *Neuron*, *36*(6), 1183–1194.

McClelland, J. L., McNaughton, B. L., & O'Reilly, R. C. (1995). Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, *102*(3), 419–457.

Schapiro, A. C., Turk-Browne, N. B., Botvinick, M. M., & Norman, K. A. (2017). Complementary learning systems within the hippocampus: A neural network modelling approach to reconciling episodic memory with statistical learning. *Philosophical Transactions of the Royal Society B*, *372*(1711), 20160049.

Spens, E., & Burgess, N. (2024). A generative model of memory construction and consolidation. *Nature Human Behaviour*, *8*(3), 526–543.

Wilson, M. A., & McNaughton, B. L. (1994). Reactivation of hippocampal ensemble memories during sleep. *Science*, *265*(5172), 676–679.

Page 2

THE DRAFT

Format	2-page CCN extended abstract, double-blind
Done	Abstract, introduction, methods and analysis plan
Figure	Architecture of the fast, slow and perception networks
Next	Fill in results, write the discussion, submit

Target venue: Conference on Cognitive Computational Neuroscience (CCN)

-------- Thank You! ---------

Long distance Python Goddess